Naive bayes

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July 2026

Naive Bayes Classifier

1 Introduction

Naive Bayes is a **supervised machine learning algorithm** used for **classification**. Unlike Decision Trees or K-Nearest Neighbors (KNN), Naive Bayes is based on **probability theory** and Bayes' Theorem.

Instead of asking:

- Which neighbor is closest?
- Which branch should be followed?

Naive Bayes asks:

> *"What is the probability that this sample belongs to each class?"*

The algorithm computes the probability of every possible class and predicts the class with the highest probability.

1.1 Example

Suppose we receive an email containing the words:

- Free
- Winner
- Money

Naive Bayes computes

$$P(Spam|Email)$$

and

$$P(NotSpam|Email)$$

If

$$P(Spam|Email) > P(NotSpam|Email),$$

then the email is classified as **Spam**.

2 Probability

Probability measures how likely an event is to occur.

$$0 \leq P(A) \leq 1$$

where

- $P(A) = 0$ means the event is impossible.
- $P(A) = 1$ means the event is certain.

2.1 Example

When rolling a fair six-sided die,

$$P(Rolling a 3) = \frac{1}{6}.$$

The probability of obtaining an even number is

$$P(Even) = \frac{3}{6} = \frac{1}{2}.$$

3 Conditional Probability

Sometimes the probability of an event depends on another event that has already occurred. This is called **conditional probability**.

It is written as

$$P(A|B),$$

which is read as:

Probability of event A given that event B has occurred.

3.1 Example

Suppose 100 students took an exam.

- 60 students studied.
- Among those 60 students, 48 passed.

Therefore,

$$P(Pass|Study) = \frac{48}{60} = 0.8.$$

This means that a student who studied has an 80% probability of passing the exam.

4 Bayes' Theorem

Naive Bayes is based entirely on **Bayes' Theorem**, which allows us to update probabilities after observing new evidence.

The theorem is

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}.$$

where

- $P(A)$ is the **prior probability**.
- $P(B|A)$ is the **likelihood**.
- $P(B)$ is the **evidence**.
- $P(A|B)$ is the **posterior probability**.

In machine learning,

- A represents the class.
- B represents the observed features.

The theorem updates our belief about a class after observing new information.

5 Why Is It Called "Naive"?

The algorithm is called **Naive** because it assumes that all features are conditionally independent given the class.

Mathematically,

$$P(x_1, x_2, \ldots, x_n|C) = P(x_1|C)P(x_2|C) \cdots P(x_n|C).$$

Although this assumption is often unrealistic, Naive Bayes performs remarkably well in many real-world applications.

5.1 Example

Suppose we classify fruits using the following features:

- Color
- Shape
- Size

Naive Bayes assumes that, once the fruit class is known, these features are independent of one another.

6 How Naive Bayes Works

Assume there are two possible classes:

- Spam
- Not Spam

For a new sample, the classifier computes

$$P(Spam|X)$$

and

$$P(NotSpam|X),$$

where X represents the observed features.
The predicted class is the one with the highest probability.
The general procedure is

1. Compute the prior probability of each class.
2. Compute the likelihood of each feature.
3. Multiply the probabilities using the independence assumption.
4. Compare the resulting probabilities.
5. Select the class with the highest probability.

7 Types of Naive Bayes

Three common versions of Naive Bayes are widely used.

7.1 Gaussian Naive Bayes

Gaussian Naive Bayes is designed for continuous numerical features.
Examples include

- Height
- Weight
- Age
- Temperature

It assumes that the features follow a Gaussian (normal) distribution.

7.2 Multinomial Naive Bayes

Multinomial Naive Bayes is designed for count data.

Typical applications include

- Word frequencies
- Number of clicks
- Document classification

It is one of the most popular algorithms for text classification.

7.3 Bernoulli Naive Bayes

Bernoulli Naive Bayes is used when features are binary.

Examples include

- Word present or absent
- Purchased or not purchased
- Clicked or not clicked

8 Worked Example

Suppose we want to determine whether a day is suitable for playing tennis.

The training data are shown below.

Weather	Play
Sunny	Yes
Sunny	Yes
Rain	No
Rain	No
Sunny	Yes

A new observation has

$$Weather = Sunny.$$

First, compute the prior probabilities.

$$P(Yes) = \frac{3}{5},$$

$$P(No) = \frac{2}{5}.$$

Next, compute the likelihoods.

$$P(Sunny|Yes) = 1,$$

$$P(Sunny|No) = 0.$$

Finally,

$$P(Yes|Sunny) \propto 1 \times \frac{3}{5} = \frac{3}{5},$$

$$P(No|Sunny) \propto 0 \times \frac{2}{5} = 0.$$

Since

$$\frac{3}{5} > 0,$$

the prediction is

$$Play = Yes.$$

9 Applications

Naive Bayes is widely used in

- Spam email filtering
- Sentiment analysis
- News classification
- Language identification
- Medical diagnosis
- Document categorization

10 Advantages

- Easy to implement.
- Fast training.
- Fast prediction.
- Performs well on small datasets.
- Excellent for text classification.
- Can handle high-dimensional data efficiently.

11 Limitations

- Assumes feature independence.
- Performance decreases when features are strongly correlated.
- May assign zero probability to unseen feature values.
- Often less accurate than more sophisticated classifiers on complex datasets.

12 Chapter Summary

Learning Type	Supervised Learning
Task	Classification
Based On	Bayes' Theorem
Main Assumption	Conditional independence of features
Common Variants	Gaussian, Multinomial, Bernoulli
Best Applications	Text classification and spam detection
Strengths	Fast, simple and effective
Weaknesses	Independence assumption may not hold